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EduOS Cameroon — Economic and Financial Analysis

Document ID	EDUOS-ECO-001
Version	1.0
Method	Cost-benefit analysis, 10-year horizon, 8% social discount rate, conservative/central/optimistic scenarios
Inputs	Baseline Data Annex (EDUOS-BDA-001); Budget/TCO (EDUOS-BUD-001)

1. The economic problem being priced

Cameroon has already made the big textbook investment: the ERSP/PAREC programme (IDA US\$175M + GPE US\$52.45M) moved the pupil-textbook ratio from 12:1 (2016) to 3 books per 2 students (2023). The question this analysis answers is: **what is it worth to protect and optimize a national textbook flow of this scale with end-to-end digital lifecycle management?**

The loss channels are not hypothetical — they are **measured in Cameroon**. The INS PETS III education survey (2019) found **~30% of the monetary value of school “minimum package” supplies lost between council dispatch and school receipt**, 3–6 month delivery delays for >80% of school heads, and 11.8% of heads reporting losses during withdrawal (BDA §3). The quantifiable channels:

- **L1 — Distribution leakage:** books procured but never reaching classrooms (diversion, mis-routing, warehouse loss).
- **L2 — Allocation mismatch:** schools over- or under-served because forecasts use stale enrolment data, forcing emergency redistribution or leaving surpluses idle.
- **L3 — Premature book death:** without condition tracking, recovery campaigns, and repair, books exit service faster than their design life.
- **L4 — Administrative burden:** thousands of staff-days consumed by manual counting, paper returns, and reconciliation at 4 administrative levels.

2. The baseline flow being protected

Annual national textbook replacement flow, central estimate:

Parameter	Value	Basis
Books in circulation (public primary target: 3 essential books × ~4M public-primary learners, plus secondary flows)	~12–14M books	BDA §2–3
Average service life	3 years (design)	FRS default
Annual replacement + growth procurement	~4.5M books/yr	circulation ÷ life
Unit cost (post-reform, flagged ESTIMATE)	2,000 FCFA	BDA §3 range 1,500–3,000
Annual textbook procurement flow	≈ 9.0 bn FCFA/yr (US\$15.7M)	derived

This is consistent in magnitude with PAREC’s 32.3 bn FCFA 2025 allocation, of which textbook provision is a principal component. **Appraisal action:** replace this derived flow with the actual textbook line from ERSP ISRs (BDA §6.2).

3. Quantified benefit streams (annual, at full operation)

#	Benefit	Conservative	Central	Optimistic	Mechanism
B1	Leakage reduction on distribution (share of flow recovered)	3% → 0.27 bn	6% → 0.54 bn	10% → 0.90 bn	per-copy/batch passports make diversion visible (FRS §5.2); PETS III measured ~30% value leakage on Cameroon school supply packages — even the optimistic case assumes recovering only a third of the measured loss rate
B2	Forecast-driven procurement efficiency (over/under-supply reduction)	3% → 0.27 bn	5% → 0.45 bn	8% → 0.72 bn	real-enrolment forecasting replaces cascade reporting (existing Ch. 3 problem analysis)
B3	Service-life extension via condition tracking + recovery campaigns (replacement demand ↓)	+0.25 yr life → 0.70 bn	+0.5 yr → 1.29 bn	+0.75 yr → 1.80 bn	annual verification (FR-NTR-12), repair workflow, return accountability
B4	Administrative time savings (18,500 schools × 12 staff-days/yr saved × 6,000 FCFA/day, valued at shadow wage)	50% capture → 0.67 bn	75% → 1.00 bn	100% → 1.33 bn	scan-based receipt/return replaces manual registers and paper returns
Total annual benefit (bn FCFA)		1.91	3.28	4.75	

Benefits ramp: 15% of full value in Y2 (pilot), 40% Y3, 70% Y4, 100% from Y5.

Deliberately not monetized (upside): learning outcomes from higher effective book availability; audit/PETS cost reduction; procurement price effects from demand transparency; the same infrastructure serving future modules (equipment/asset passports, inspection) at near-zero mar-

ginal registry cost.

4. Results

10-year NPV at 8% discount, against total programme cost (11.62 bn FCFA over 5 years + 1.22 bn/yr recurrent thereafter):

Scenario	NPV (bn FCFA)	BCR	EIRR	Payback
Conservative	+1.6	1.11	~11%	Year 9
Central	+9.5	1.65	~22%	Year 6
Optimistic	+18.1	2.24	~33%	Year 5

Even the conservative case — 3% leakage recovery, 3% forecast gains, a quarter-year of book life, half the admin savings — clears the hurdle rate. The central case returns roughly **1.65 FCFA per FCFA invested** in quantified savings alone, before any learning-outcome value.

5. Sensitivity

Variable	Swing tested	NPV impact (central case)
Textbook unit cost 1,500 vs 3,000 FCFA	±33% on flow	NPV 5.6 → 13.4 bn (stays positive)
Benefits ramp delayed 1 year	—	NPV -1.9 bn
Capex overrun +30% (class-3 upper bound)	—	NPV -2.6 bn
B3 (book life) fails entirely	remove largest stream	NPV +3.1 bn (still positive)
Discount rate 12%	—	NPV +6.2 bn

Break-even condition: the programme is NPV-positive if it recovers just **≈ 2.1% of the annual textbook flow in combined efficiency terms** — against a **measured 30% value-leakage rate** on comparable school-supply flows in Cameroon (PETS III). The break-even threshold is one-fourteenth of the documented loss rate.

6. Fiscal sustainability

Steady-state recurrent cost (1.22 bn FCFA/yr) equals **0.13% of the two ministries' combined 2026 budgets** and **~13% of the annual textbook flow it protects**. A system costing 13% of a flow to eliminate losses plausibly $\geq 15\text{--}25\%$ of that flow is a defensible recurrent commitment; the MTEF inclusion path is specified in EDUOS-BUD-001 §6 and Risk R5.

7. Distributional note

Benefits concentrate where losses concentrate: rural and crisis-affected schools with 30:1 pre-reform ratios are precisely where distribution visibility and recovery campaigns bind. The offline-first design (BDA §5 constraints) is what makes the benefit reach the 26%-electrified rural areas rather than only connected towns — equity is engineered in, not assumed.

8. Appraisal-stage refinements

1. ~~Extract PETS III leakage rates~~ **DONE** — 30% minimum-package value leakage extracted from the report (BDA §3); remaining task is validating its applicability to textbook flows specifically during the pilot (textbooks may leak less than fungible supplies — the model's 6% central case already assumes only one-fifth of the PETS rate).
2. Substitute derived textbook flow (§2) with the exact ERSP textbook-component spend from ISRs; unit-cost input now anchored at the verified US\$2.90 (BDA §3) — the model's 2,000 FCFA input is within 15% of the verified 1,750 FCFA average and conservative on the sensitivity range already tested (§5).
3. Validate admin time savings (B4) with a 20-school time-and-motion study during the pilot — an M&E deliverable (EDUOS-MEF-001 §5).
4. Re-run this model with pilot-measured values at the Phase II gate; the phased structure means the investment decision is re-testable before 60% of funds commit.